ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 6

Sports Nutrition Fundamentals

Duration	55 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: Peri-Workout Nutrition Strategy

Fuelling the Machine: Before, During, and After

Sports nutrition is the application of nutritional science to optimise athletic performance, recovery, and adaptation. The peri-workout period (before, during, and after training) is the highest-leverage nutritional window for athletes.

Pre-Workout Nutrition (1–3 hours before)

Goal	Macronutrient	Amount	Food Examples
Top up glycogen	Carbohydrates	1–2g/kg BW	Rice, oats, roti, banana, sweet potato
Provide amino acids	Protein	20–30g complete protein	Chicken, fish, Greek yoghurt, paneer, whey
Minimise GI distress	Low fat, low fibre	Fat <15g, fibre <5g	Avoid deep-fried, high-fat, large fibre loads
Hydration	Water + electrolytes	500ml 2h before; 250ml 15min before	Water, coconut water; dilute sports drink

30-minute pre-workout option: When time is limited, a fast-digesting protein + high-GI carb combination can be consumed 30 minutes before: 25g whey + 1 banana + water. This provides amino acids and quick glucose without GI discomfort.

Intra-Workout Nutrition (during sessions >60 min)

Session Duration	Carbohydrates	Electrolytes	Notes
<60 minutes	Not needed	Water sufficient	Muscle glycogen adequate for <60 min moderate intens
60–90 minutes	15–30g/hour	Sodium 200–500mg/hour if sweating	Sports drink (6–8% carb) or banana half
>90 minutes	30–60g/hour	Sodium + potassium + magnesium	Multiple carb sources (maltodextrin + fructose) empties gut faster
Endurance events >2.5h	60–90g/hour	Full electrolyte replacement	Requires multi-transporter carb mix (glucose + fructose

Multi-transporter carbohydrates: The small intestine has two separate carbohydrate transporters: SGLT1 (glucose, maltose) and GLUT5 (fructose). Using both simultaneously allows absorption of 60–90g carbs/hour vs 60g maximum for glucose alone. This is why most endurance sports products combine maltodextrin (glucose) with fructose at a 2:1 ratio.

Post-Workout Nutrition (within 0–2 hours)

Post-workout nutrition serves two primary purposes: initiate muscle protein synthesis (MPS) and replenish muscle glycogen. These require different nutrients:

Goal	Macronutrient	Amount	Timing	Best Foods

MPS initiation	Complete protein (leucine-rich)	25–40g	Immediately post (if fasted); within 2h if pre-workout protein eaten	Whey, chicken, eggs, fish, milk, casein
Glycogen resynthesis	High-GI carbohydrates	0.8–1.2g/kg BW	ASAP post-workout (if next session within 8h)	White rice, white potato, sports drink, banana, dat
Hydration	Water + sodium	1.5L per kg bodyweight lost	During + post-workout	Water + pinch of salt; coconut water; electrolyte dr

SECTION 2: Nutrition for Different Sports Demands

Sport-Specific Nutritional Demands

Sport Type	Energy System	Primary Fuel	Carb Need	Protein Need
Power/Strength (Olympic lifting, powerlifting)	ATP-PCr → glycolytic	PCr, glycogen	Moderate (4–6g/kg)	High (1.8–2.2g/kg)
Bodybuilding	Glycolytic (moderate)	Glycogen	Moderate-high (4–7g/kg)	High (1.8–2.5g/kg)
HIIT / CrossFit	Glycolytic + oxidative	Glycogen + fat	High (5–7g/kg on training days)	High (1.8–2.2g/kg)
Endurance running (marathon)	Oxidative	Fat + glycogen (fat majority)	Very high (7–10g/kg day before)	Moderate (1.4–1.7g/kg)
Team sport (football, basketball)	Mixed all three	All substrates	High (6–8g/kg)	Moderate-high (1.6–2.0g/kg)
Combat sport (boxing, wrestling)	Glycolytic dominant	Glycogen, PCr	Moderate-high (5–7g/kg)	High (1.8–2.5g/kg)

SECTION 3: Weight Cutting and Making Weight (Combat Sports Context)

Rapid Weight Loss for Competition

Athletes in weight-class sports (boxing, wrestling, judo, powerlifting) routinely cut weight before weigh-in. Understanding the physiology is critical for both practitioners and coaches:

Weight Cut Method	Weight Lost	Recovery Time Needed	Risks
Caloric restriction (4–8 weeks)	Fat mass	N/A — permanent loss (fat)	Muscle loss if insufficient protein; reduced strength
Water manipulation (24–48h)	Water + electrolytes (2–5kg)	24h required for rehydration	Impaired performance; cramping; dangerous if >5% BW
Glycogen depletion	Glycogen + water (1–3kg)	24–48h carb reload needed	Performance impairment if not refuelled
Sweat suit / sauna (acute)	Water (1–3kg)	24–48h rehydration	Dangerous; banned in some sports; cardiac risk

Safe weight class strategy: Compete as close to walk-around weight as possible. Chronic severe weight cutting (>5% body weight) impairs long-term health, bone density, and hormonal function. Use caloric restriction only to reduce fat mass over the competition preparation period.

SECTION 4: Hydration Science for Athletes

Advanced Hydration for Athletic Performance

Elite athletes use sweat rate testing to individualise hydration strategies. Sweat rates vary from 0.5L/hr (cold weather, low intensity) to 3L/hr (heat, high intensity). Sodium concentration in sweat also varies (300–1800mg/L) — the highest sodium sweaters are at greatest risk of exercise-associated hyponatraemia when over-hydrating with plain water.

Hydration Metric	Test Method	Implications for Training
Sweat rate	Weigh before/after: (pre-BW – post-BW + fluids consumed) / time	Target to replace 75–80% of fluid losses during exercise
Sweat sodium	Commercial sweat patch (Nix Biosensor, FLOW)	High sodium sweaters need electrolyte drinks even in short sessio
Urine specific gravity	Refractometer; home test: first void USG >1.020 = dehydrated	Useful for pre-competition hydration monitoring

Module 6 (CS_B2) Key Takeaways

- Pre-workout: 1–2g carbs/kg + 20–30g protein + minimal fat/fibre 1–3h before training
- Intra-workout carbs only needed for sessions >60 minutes; 30–60g/hour using multi-transporter carbs
- Post-workout: 25–40g complete protein + high-GI carbs within 2 hours
- Sport-specific carbohydrate needs vary dramatically: powerlifters need 4–6g/kg; marathon runners 7–10g/kg
- Weight cutting should primarily use fat loss over weeks, not acute water manipulation
- Individual sweat rate and sodium concentration testing allows precision hydration programming

1. Explain why multi-transporter carbohydrates (maltodextrin + fructose) allow higher absorption rates than glucose alone.

2. Compare peri-workout nutrition for a powerlifter vs a marathon runner.

3. What is sweat rate, how is it calculated, and how does it influence hydration strategy?

4. Why is acute severe weight cutting (>5% body weight) dangerous, and what is the evidence-based alternative?

5. What post-workout nutrition protocol would you recommend for an athlete training twice per day?